Exercises

1a.

0 Order: $P0\ (x)=-62$

1st Order: $P1\ (x)=-62+70(x-1)$

2nd Order: $P2\ (x)=-62+70(x-1)+69(x-1)^2$

3rd Order: $P3\ (x)=-62+70(x-1)+69(x-1)2+25(x-1)^3$

1b

0: 0.6931

1st: 1.6931

2nd: 1.931

3rd:1.5264

2 Yes, Newton's method can be used, but the convergence will be Linear.

3 False. We only know that f(x*) = 0 and nothing about adding alpha to the equation.

Computer Lab

1 root3 = 1.732 | 5root7 = 1.476

2a

First derivative: $h'(x)=sec2(x)-1=tan2(x)$.

Second derivative: $h''(x)=2tan(x)sec2(x)$.

2b

4.96 because the the function tan has a vertical asymptote at x = 3pi/2

2d

4.715

AI Prompt used for computer lab (I couldn’t fix the formatting)

Write a complete, clean MATLAB script using Newton's method to solve two problems. First, implement an nth

root finder to calculate 3 and 71/5

. Second, find the root of h(x)=tan(x)−x near 4.5 by performing iterations from x0

=4.2 and x0

=4.7, plotting the convergence error and error ratio, and calculating the theoretical prefactor |h''

(x∗

)/2h'

(x∗

)|. Repeat the convergence plot and prefactor calculation for the alternative formulation g(x)=xcos(x)−sin(x).